

HIA PAUL

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PROFESSIONAL SUMMARY

Data Science Intern at Shabd Intelligence Pvt Ltd with hands-on experience in deploying proprietary LLMs on Linux servers, implementing RAG pipelines with vector databases, and building production-grade chatbots and document QA systems. Skilled in fine-tuning transformer models, API integration, prompt engineering, and delivering client-facing demos. Passionate about uncovering insights and trends in data. **CGPA: 9.43**. Strong foundation in Machine Learning, Deep Learning, and Linux system administration.

EDUCATION

B.Tech in CSE (Data Science) <i>Brainware University</i>	CGPA: 9.43	2022 - 2026
Higher Secondary (WBHSE) <i>Patha Bhavan, Kolkata</i>	75%	2020 - 2022
Secondary (WBSE) <i>Brahma Balika Shikshalaya, Kolkata</i>	85%	2018 - 2020

TECHNICAL SKILLS

LLMs & NLP	Proprietary LLMs, Hugging Face Transformers (BERT, RoBERTa, GPT, LLaMA), OpenAI API, Gemini API, Fine-tuning, Tokenization, NER, Text Classification, Prompt Engineering, RAG, Speech Data Preprocessing
Vector Databases	FAISS, ChromaDB, Pinecone, Vector Embeddings, Similarity Search, Document Chunking
ML & Deep Learning	Regression, Classification, Clustering, CNN, Neural Networks, Transfer Learning (VGG16, ResNet, MobileNet), Model Evaluation
Programming	Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, LangChain
APIs & Web	FastAPI, Flask, RESTful API Development, API Key Management, Postman, cURL, Streamlit, Gradio
Databases	MySQL, SQL, Vector Databases
Linux & DevOps	Linux Server Administration, Bash Scripting, Cron Jobs, Git, GitHub, Docker, MLflow, Conda/Venv
Data Visualization	Power BI, Tableau, Matplotlib, Seaborn
Soft Skills	Problem Solving, Technical Documentation, Client Demos, Cross-functional Collaboration, Core Banking Implementation

EXPERIENCE

DATA SCIENCE INTERN		
Shabd Intelligence Pvt Ltd	Kolkata, WB	April 2026 - Present
- Deployed Shabd's proprietary/personalized LLM on Linux servers for production-grade applications including chatbots and document QA systems. - Contributed to AI-powered speech processing for Indian languages through speech data preprocessing, NLP tasks, and multilingual dataset handling. - Integrated external LLM APIs (OpenAI, Hugging Face, Gemini) for hybrid model approaches and benchmarking against proprietary models. - Implemented prompt engineering techniques (few-shot, chain-of-thought, system prompts) to optimize domain-specific responses, reducing hallucinations by 30%. - Built Retrieval-Augmented Generation (RAG) pipelines using FAISS/ChromaDB vector databases for context-aware responses from company documents and knowledge bases. - Designed document chunking and embedding strategies for efficient similarity search, reducing retrieval latency by 35%.		

- Developed **interactive chatbots** and **document QA systems** using proprietary LLM and RAG pipelines.
- Created **client-facing demo interfaces** using Streamlit/Gradio for showcasing AI capabilities.
- Delivered live demos to internal stakeholders and potential clients, demonstrating business value of AI solutions.
- Built **RESTful APIs** using FastAPI to serve LLM models, RAG pipelines, and chatbots with asynchronous processing.
- Managed API keys and authentication securely using environment variables and secrets management.
- Managed Linux server environments including package installation, dependency management, virtual environment setup (conda/venv), and cron job scheduling.
- Monitored GPU utilization using nvidia-smi and optimized inference latency by **40%** using ONNX/TorchScript conversions.
- Wrote Bash scripts for automating data backup, log rotation, model deployment, and scheduled retraining.
- Implemented model versioning and experiment tracking using **Git** and **MLflow**.
- Wrote complex **SQL queries** for data extraction, cleaning, and preprocessing.
- Collaborated with AI team on end-to-end NLP data processing pipelines using Pandas and Hugging Face datasets.

DATA SCIENCE INTERN

Euphoria GenX

Kolkata, WB

August 2025 - November 2025

- Developed **Mobile Usage Dashboard** analyzing user mobile activity and usage patterns.
- Collected, cleaned, and processed datasets using **Python, Pandas, and Excel**.
- Performed data analysis to identify trends in app usage and screen time behavior.
- Built interactive dashboards and visual reports using **Power BI/Tableau** to present insights effectively.
- Collaborated with team members on project implementation while enhancing skills in data analysis, visualization, and problem-solving.

PROJECTS

BRAIN TUMOR DETECTION USING DEEP LEARNING

Python — PyTorch — TensorFlow

- Implemented CNN and transfer learning models (VGG16, ResNet50, MobileNetV2, EfficientNetB0) for brain tumor classification.
- Performed comparative analysis using accuracy, precision, recall, and F1-score metrics.
- Achieved **92% accuracy** with EfficientNetB0 on test dataset.

HOUSE PRICE PREDICTION USING MACHINE LEARNING

Python — Scikit-learn — Pandas — Matplotlib

- Built regression model with comprehensive feature engineering (handling multicollinearity, outlier detection, missing data imputation).
- Applied algorithms: Linear Regression, Decision Trees, Random Forest, XGBoost.
- Achieved **R² score of 0.89** on real-world housing dataset.

KEY ACHIEVEMENTS

- Successfully deployed **Shabd's in-house LLM** for chatbot & document QA in production.
- Built **RAG pipelines** using FAISS/ChromaDB for context-aware document retrieval.
- Developed and presented **interactive AI demos** to stakeholders & clients.
- Reduced LLM hallucinations by **30%** via advanced prompt engineering.
- Reduced retrieval latency by **35%** and inference latency by **40%**.
- Achieved **92% F1-score** on custom datasets using Hugging Face models.
- Active participant and Lead Presenter for multiple college hackathons.

CERTIFICATIONS & ACHIEVEMENTS

- **The Fledgling** - Certification
- **Active Participant** in college hackathons; Lead Presenter pitching solutions to industry judges.
- **CGPA: 9.43** - Consistent academic excellence throughout B.Tech.
- Hands-on experience with **proprietary LLMs, RAG, and vector databases**.
- Music enthusiast (singing) and creative cook.